Week :9 (26-30 October 2026)

من – إلى الأسبوع #:

Subject/الموضوع: Physics **Unit Title/الوحدة عنوان: Standing waves** **Grade/: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 قانون بويل	I can explain how standing waves are formed and use mathematical relationships to determine the wavelength, frequency, and harmonic modes of standing waves in different systems.	2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 2 الحصة الثانية	I can explain phase differences in standing waves and use them to describe the motion of particles at different points along a standing wave.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 3 الحصة الثالثة	I can explain resonance and damping and describe their effects on the amplitude of oscillating systems.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	please check student Teams
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a charged devices, calculator, Physics notebook, ruler, pencil daily, participate actively, review feedback and upload required work to Toddle and Teams.		

Week # 4 الأسبوع#

Technology Integration دمج التكنولوجيا	Digital graphing or teacher-provided graph displays; calculators for gradients, areas and equations; motion videos or simulations where available; Teams for resources, feedback and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: gradient, geometric area, algebra, rearranging equations, proportional reasoning and negative values. English: interpret motion descriptions and justify conclusions. UAE engineering and road safety: analyse transport data and stopping motion.
External Assessment Integration دمج التقييمات الخارجية	NGSS-style external assessment questions requiring students to analyse data, interpret multi-section graphs, calculate gradient and signed area, select kinematics equations, translate between representations and justify conclusions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking: analyse patterns, connect representations and evaluate solutions. Communication: explain graph features and justify equation choices. Self-management: organise multi-step working and check signs and units. Collaboration: critique reasoning respectfully. Wellbeing: build confidence through scaffolded practice and purposeful challenge.
Standards المعايير	HS-PS2-1: Analyze data to support the claim that Newtons second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Supporting alignment: students analyse motion data and graphical acceleration as the kinematic foundation for later force-motion modelling.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	How can the same journey be represented by equations and different motion graphs, and how can we verify that the representations are consistent?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1** **Must: I must identify nodes and antinodes on a standing wave.** **Should: I should explain how standing waves are formed through the superposition of two waves traveling in opposite directions.** **Could: I could solve problems involving harmonics, wavelength, and frequency in standing wave systems.** . **Lesson 2:** **Must: I must identify which points on a standing wave oscillate in phase and which oscillate in antiphase.** **Should: I should explain why all particles between two adjacent nodes oscillate in phase.** **Could: I could analyze standing-wave diagrams to determine the phase relationship between any two points on the wave.** **Lesson3:** **Must: I must define resonance and damping and describe their effects on oscillations.** **Should: I should explain how the driving frequency affects the amplitude of a system and why resonance occurs at the natural frequency.** **Could: I could analyze real-world examples of resonance and damping and evaluate their benefits or risks in practical applications.**
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding العال م التفاهم History التاري••••خ **Area:** UAE Transportation, Road Safety, Innovation and Sustainable Infrastructure المجال Implementation: Students analyse Abu Dhabi transport motion using graphs, gradients, signed areas and kinematics equations, connecting accurate modelling to UAE road safety and smart mobility. التطبيق